

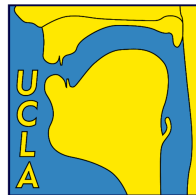
Modeling Vowel System Typology Using Iterated Confusion Minimization



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Dispersion Theory

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- e.g. 'corner' vowels /iau/ found in most systems minimize confusion
- Perceptual contrast optimized using 'energy' function based on pairwise psychoacoustic distance between vowels
- Optimal systems match typology, e.g. /iau/, /ieau/, /ieaou/, etc.

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- Adaptive Dispersion Theory (Lindblom, 1986)
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- OT-Dispersion Theory (Flemming, 1995, 2004)

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Schwartz et al. (1997b) provides a benchmark for performance of acoustically-based dispersion models in predicting typologically dominant vowel systems

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- Lack of mechanism for system emergence

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- Both accounts involve incremental adjustments to phonetic distributions that reduce confusability between phonetic categories

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- ① Quantitative measure of perceptual contrast
- ② Prediction of typological frequencies
- ③ Mechanism compatible sound change

Iterated Confusion Minimization

The key? **Iterated Confusion Minimization (ICM)**

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ICM simulates mathematically optimal listeners and speakers

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Simplifying assumption: vowel categories have equal prior probabilities

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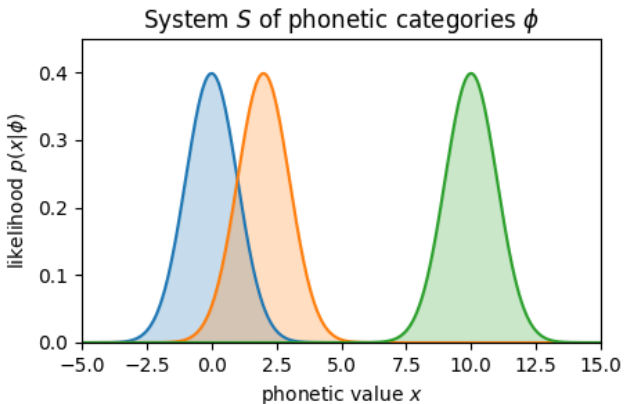
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Simplifying assumption: vowel categories have equal prior probabilities

⇒ Optimal Bayesian listener follows **maximum-likelihood behavior**

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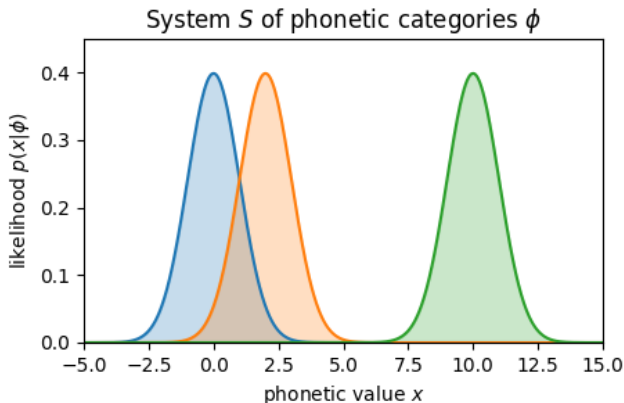


Optimal Listener Behavior

Tokens classified as category with highest likelihood at phonetic value

maximum-likelihood behavior

$$C(x) = \operatorname{argmax}_{\phi \in S} p(x|\phi)$$

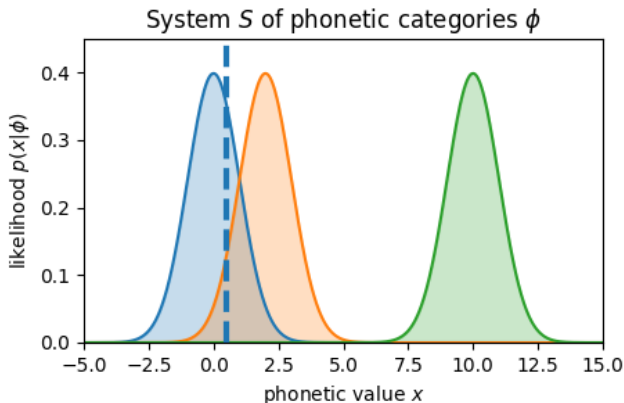


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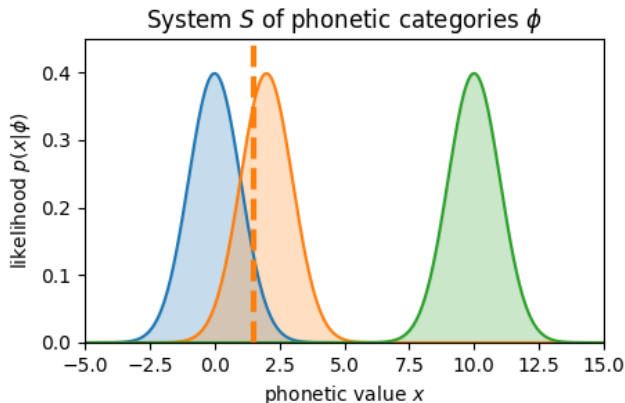


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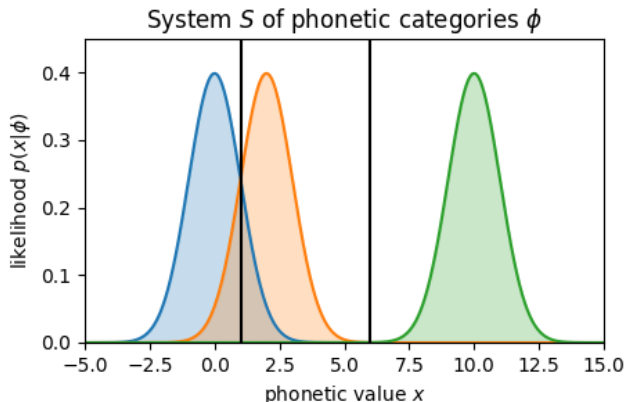
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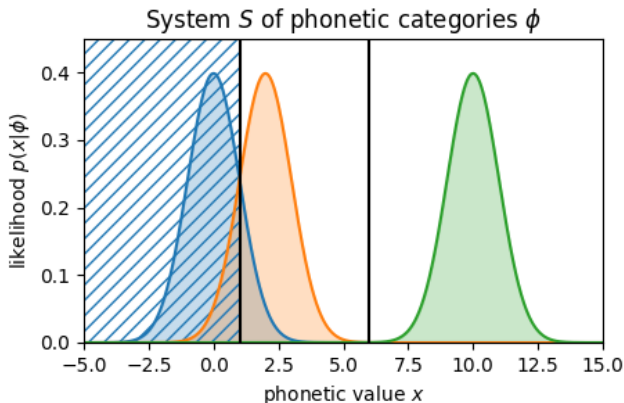
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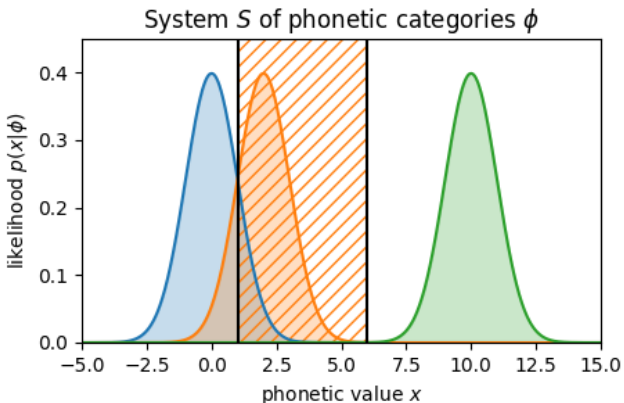
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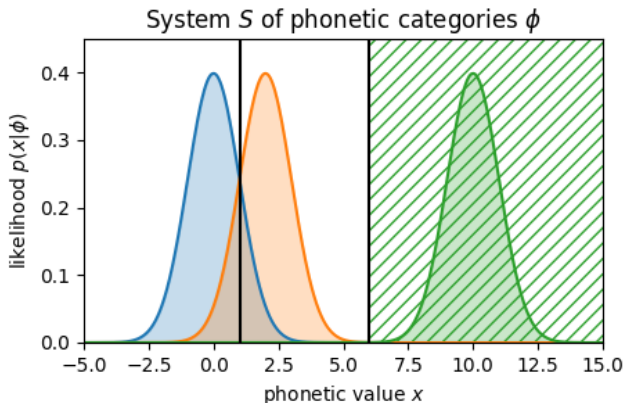
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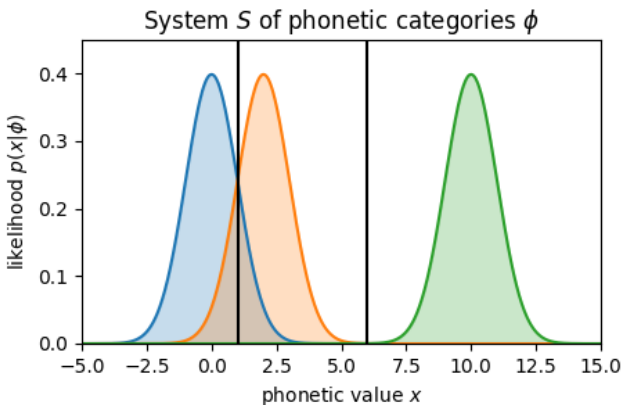
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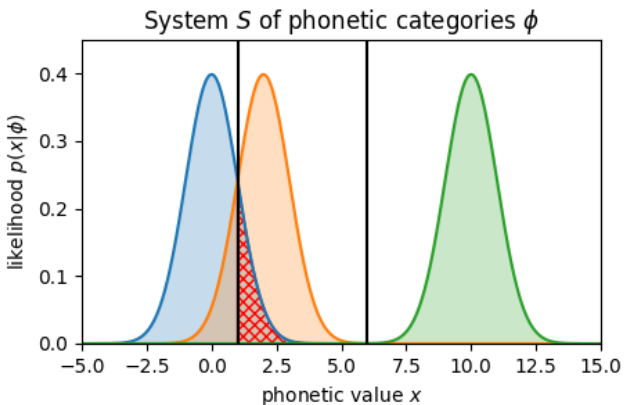
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Tokens on 'wrong' side of boundaries
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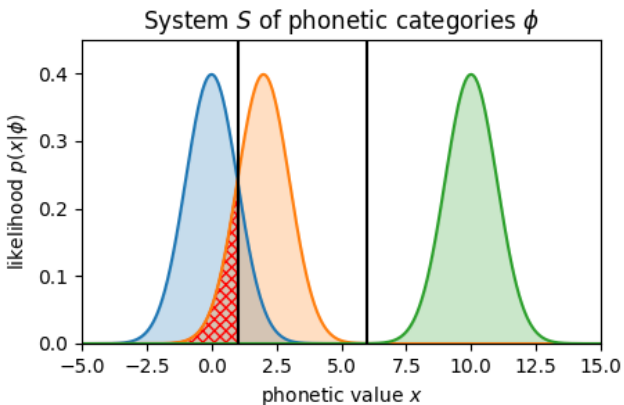
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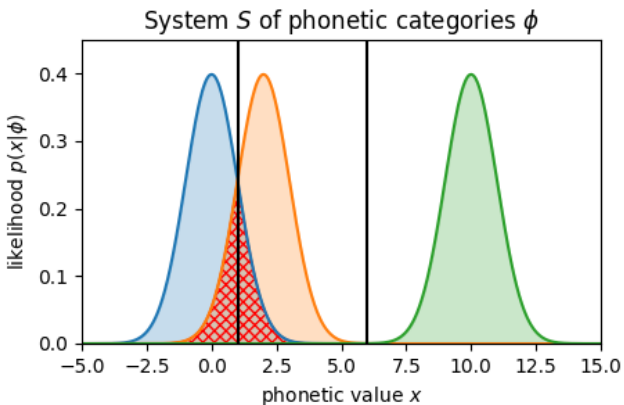
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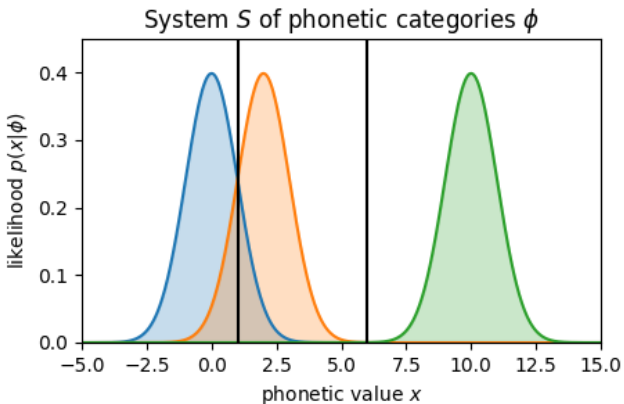
Confusion is equivalent to area of phonetic overlap (times normalizing constant)

classifier confusion rate

$$\frac{1}{|S|} \sum_{\phi \in S} \int p(x|\phi) P(C(x) \neq \phi) dx$$



Optimal Speaker Behavior



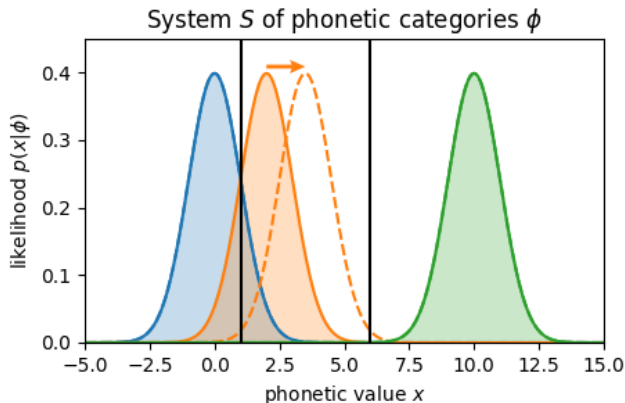
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A new set of distributions can reduce confusion, even if the listener expects the original distributions

least confusing system S'

Given set of possible distributions Φ :

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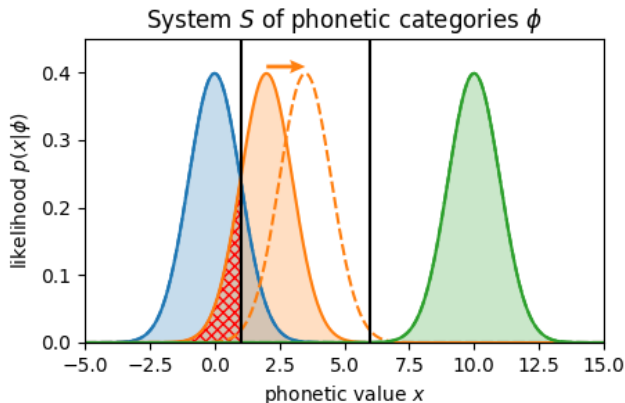
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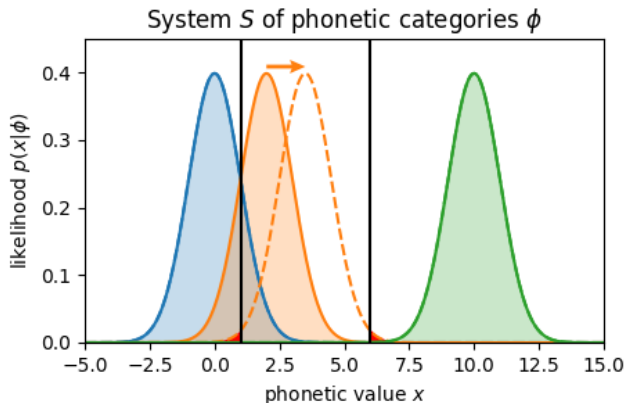
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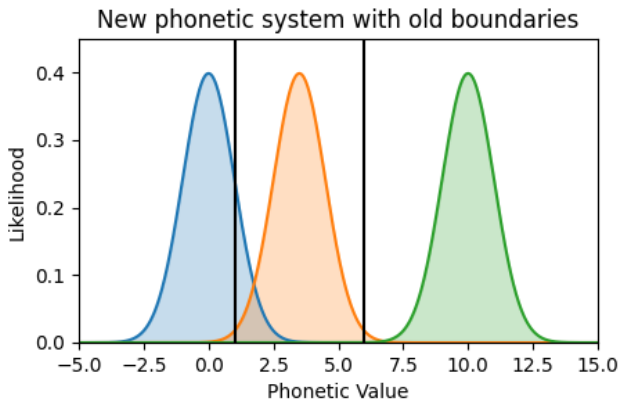
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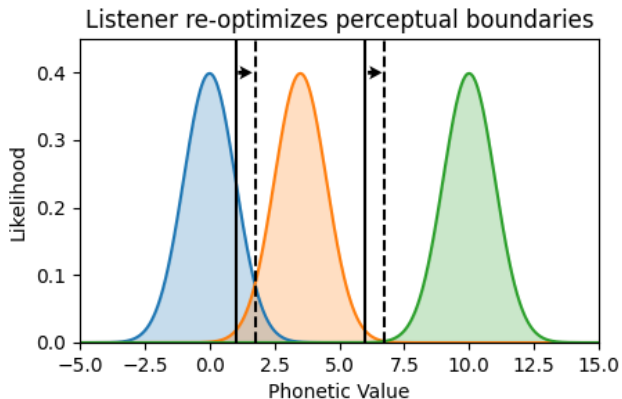


Iterated Confusion Minimization



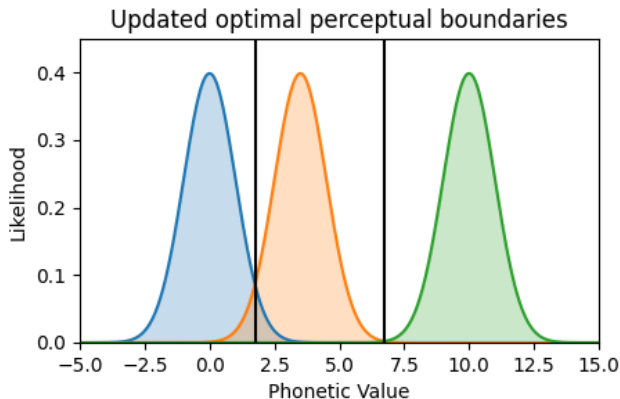
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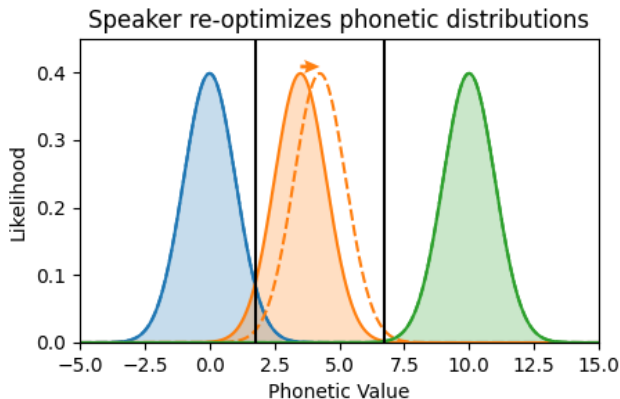
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Given speaker's new system, the optimal listener updates their perceptual boundaries

Then, the optimal speaker updates their phonetic distributions again

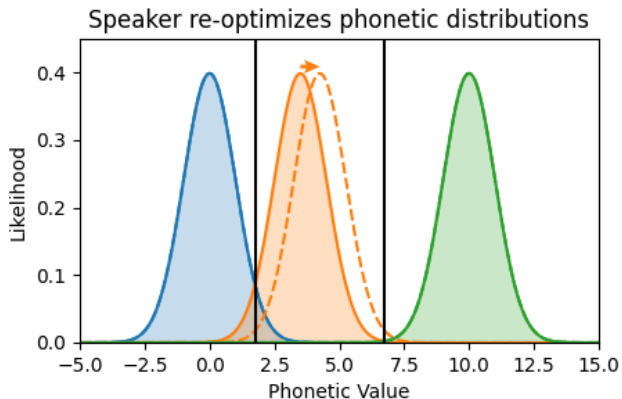


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Given speaker's new system, the optimal listener updates their perceptual boundaries

Then, the optimal speaker updates their phonetic distributions again

We will call this repeated process
Iterated Confusion Minimization (ICM)



Iterated Confusion Minimization Algorithm

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Input: $S_0 \subseteq \Phi$, where Φ is set of possible phonetic distributions

Output: $S_i \subseteq \Phi$

1: $i \leftarrow 0$

2: **repeat**

3: ▷ *Optimal listener behavior*

4: $C_i(x) \leftarrow \operatorname{argmax}_{\phi \in S_i} p(x|\phi)$

5: ▷ *Optimal speaker behavior*

6: $S_{i+1} \leftarrow \{\operatorname{argmin}_{\phi' \in \Phi} P(C_i(\phi') \neq \phi) | \phi \in S_i\}$

7: $i \leftarrow i + 1$

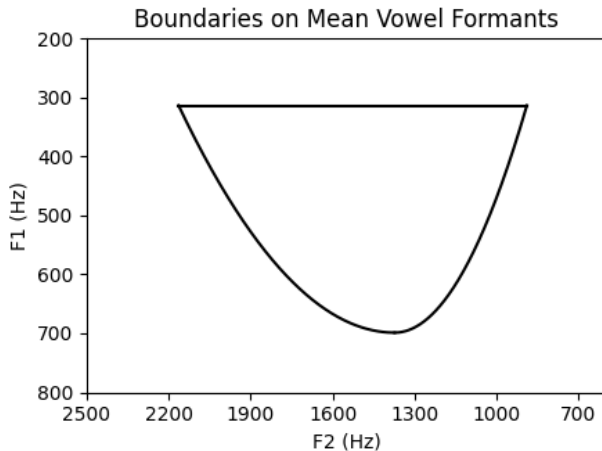
8: **until** $S_i = S_{i-1}$

9: **return** S_i

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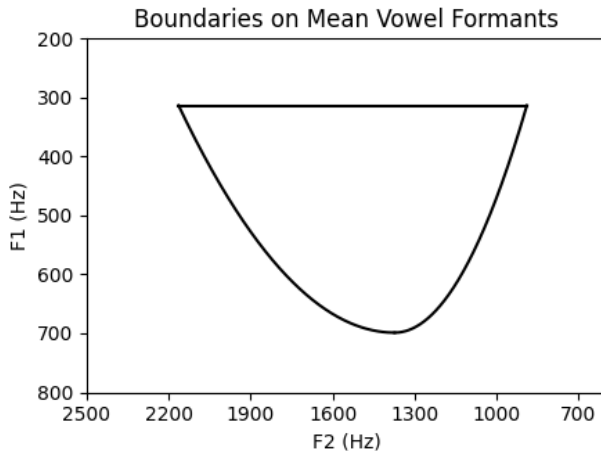
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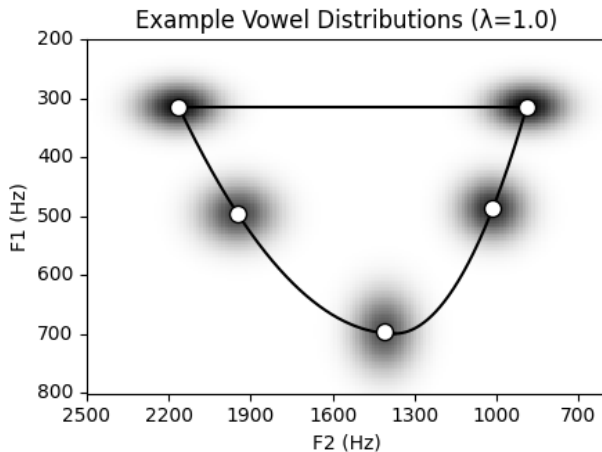
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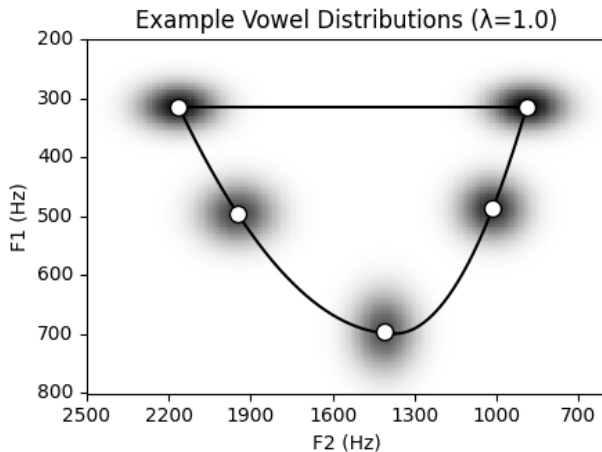


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F2 standard deviations scaled using
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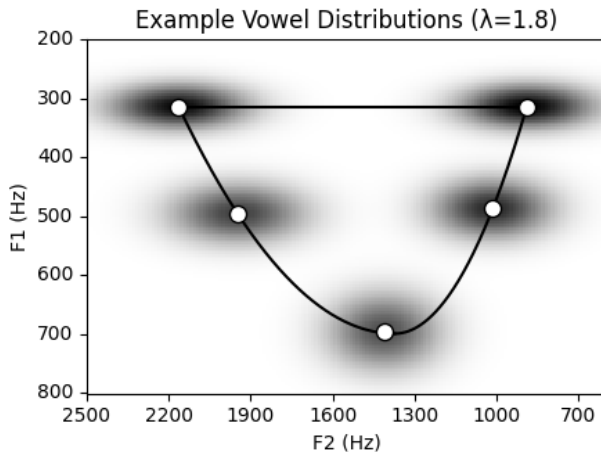


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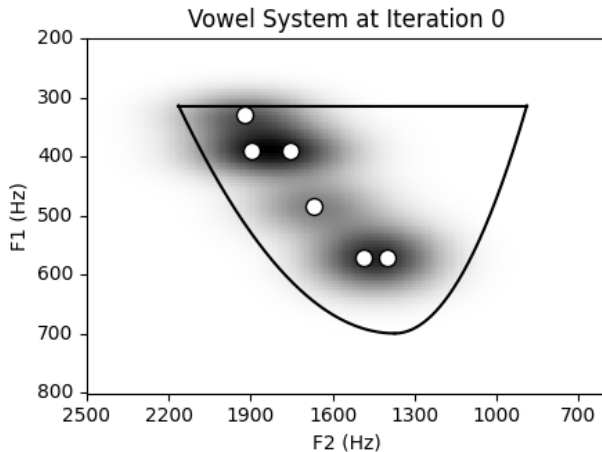
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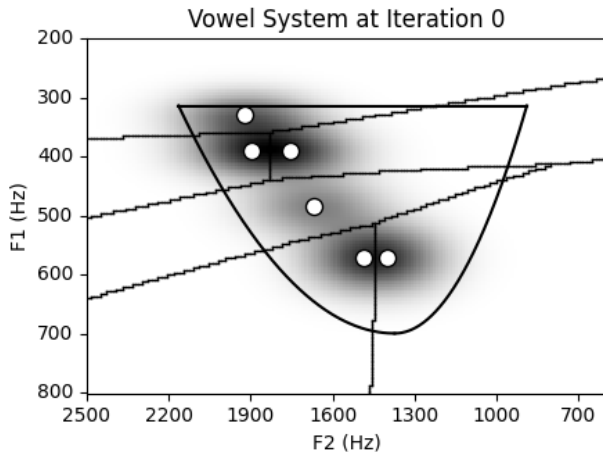
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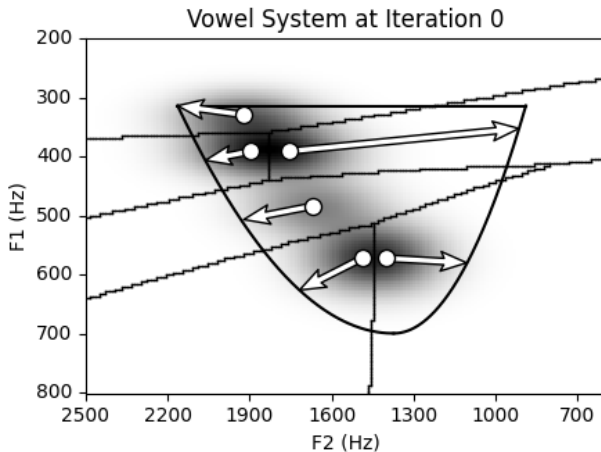


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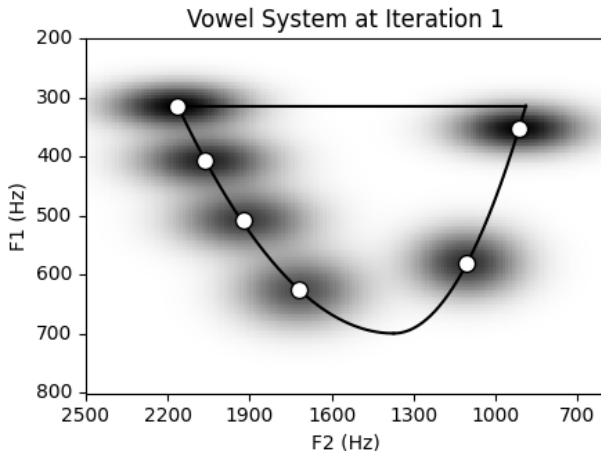
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ICM exerts strong pressure towards vowel space periphery



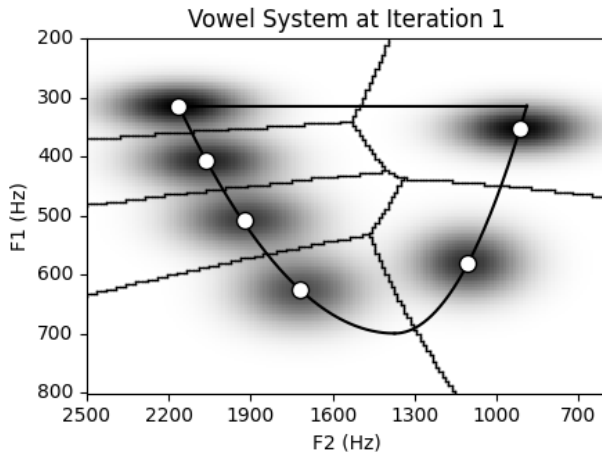
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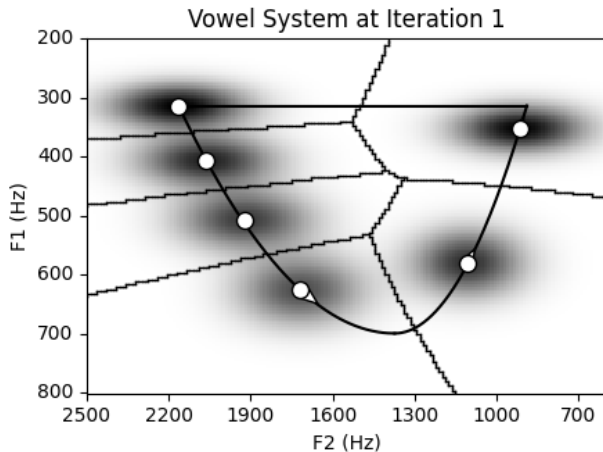


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Simulation yields realistic chain shift along lower periphery

- cf. California shift, Canada shift

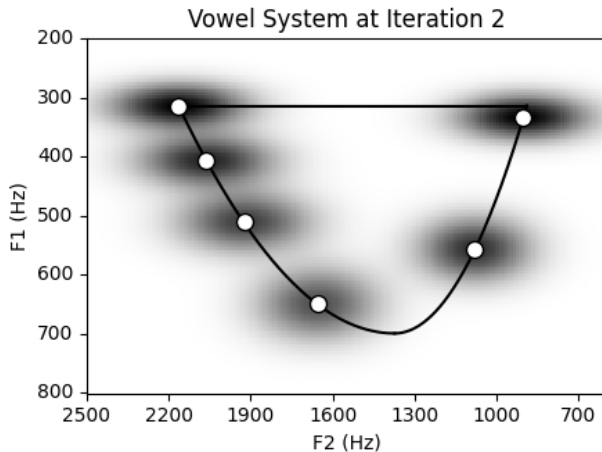


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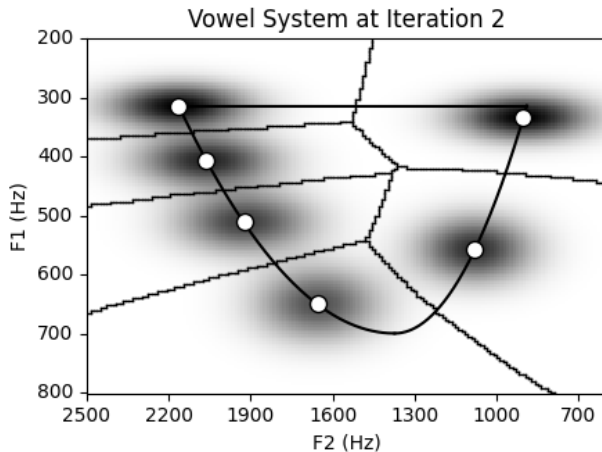


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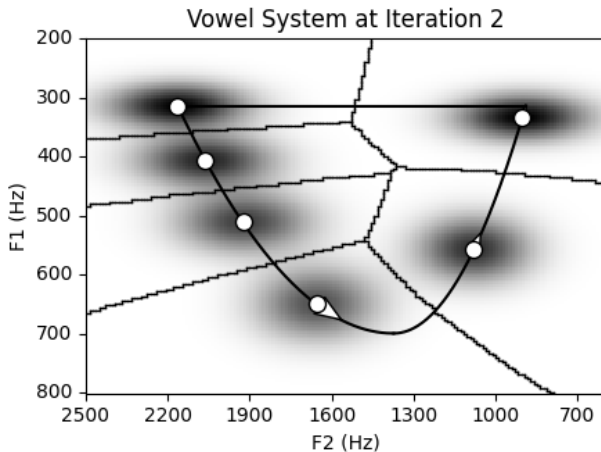


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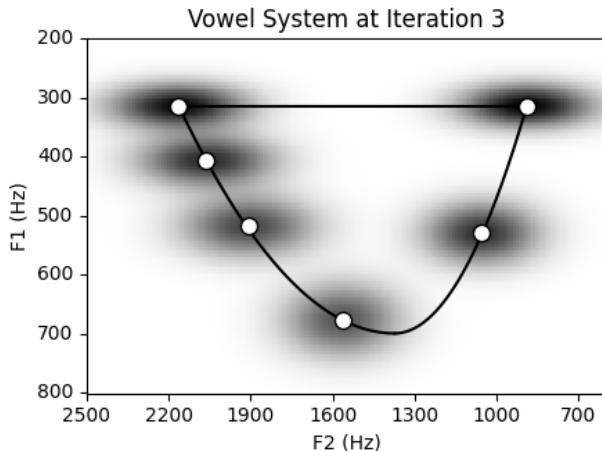


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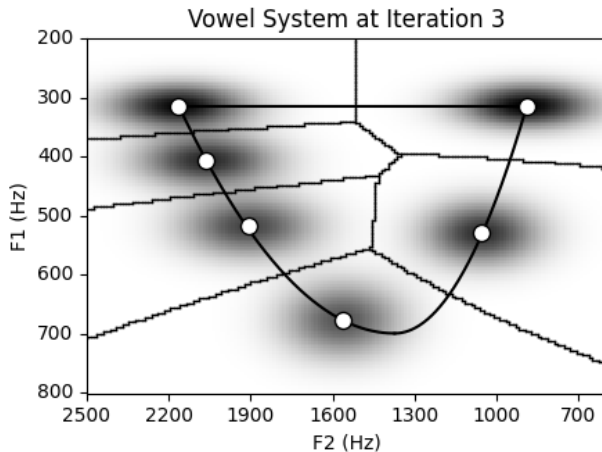


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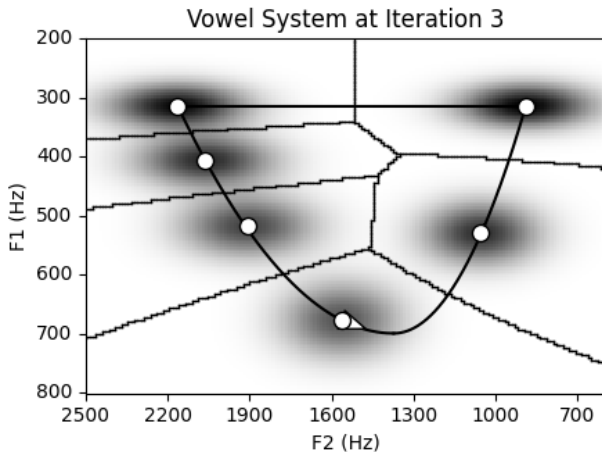


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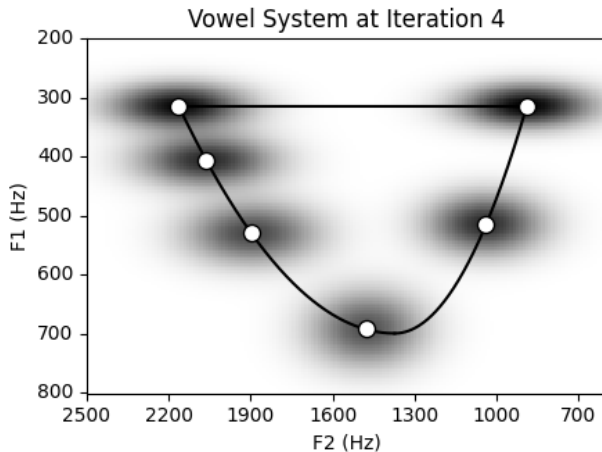


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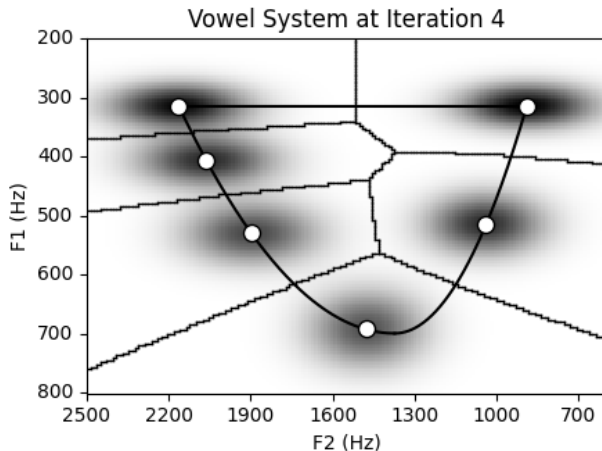


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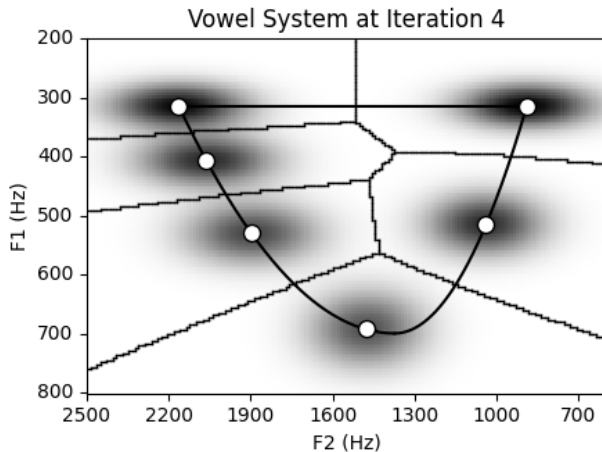


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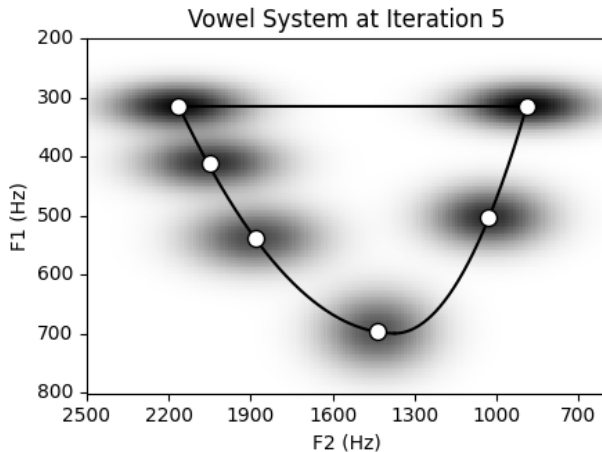


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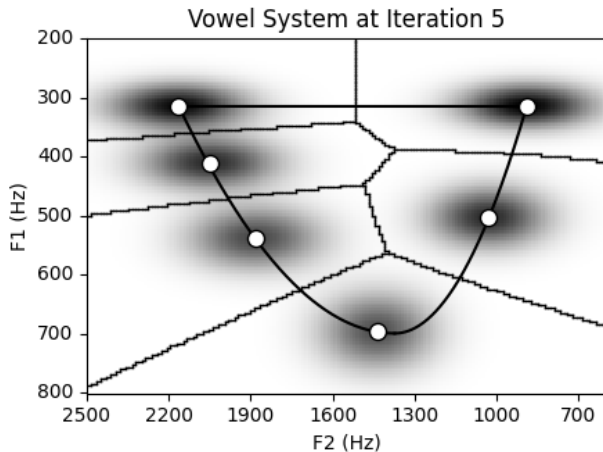


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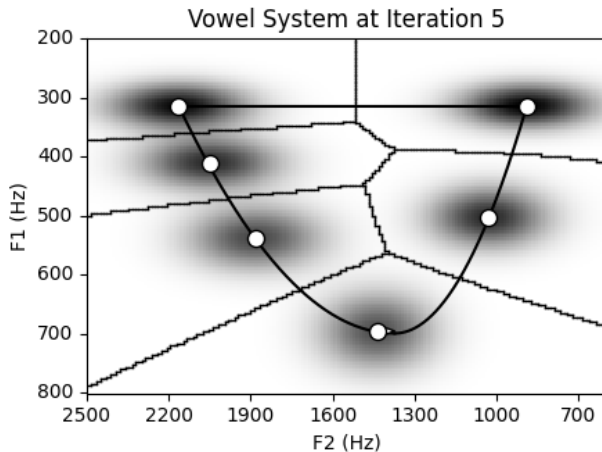


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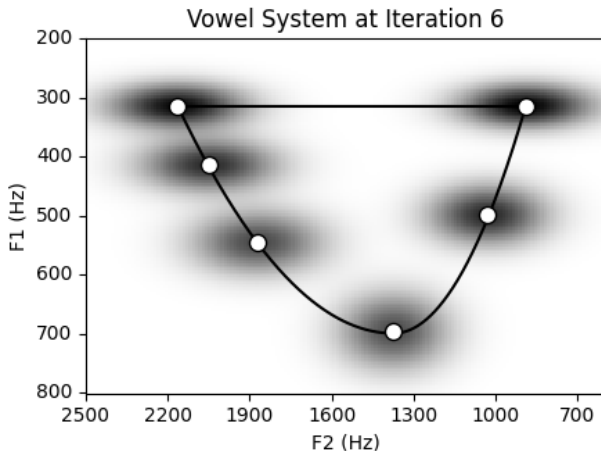


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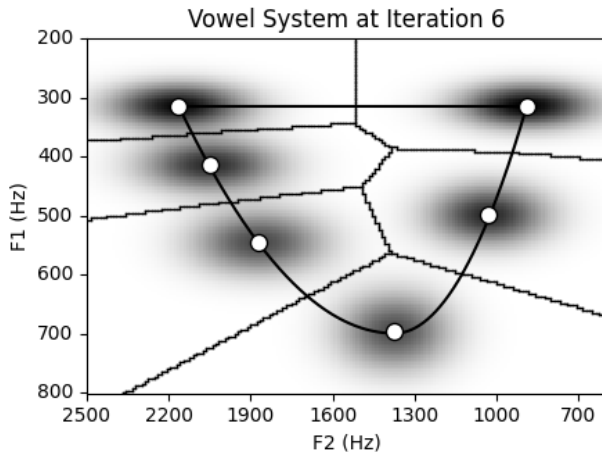


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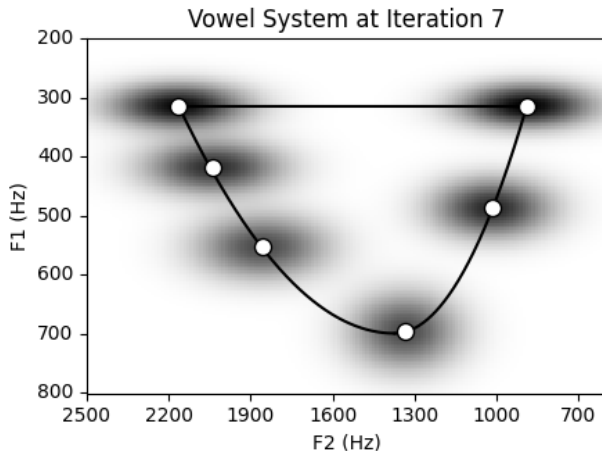


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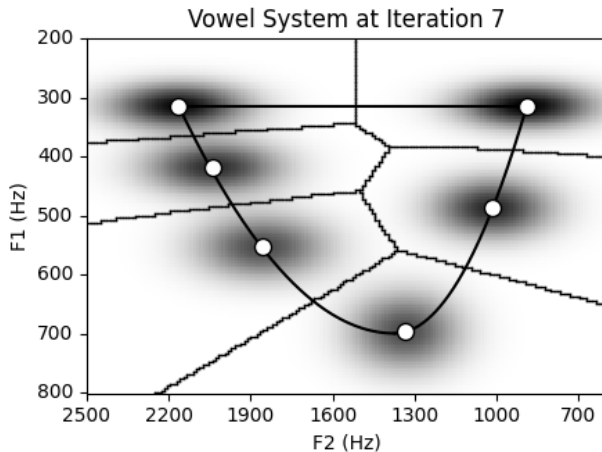


Example Result: Predicting Chain Shifts

ICM exerts strong pressure towards vowel space periphery

Simulation yields realistic chain shift along lower periphery

- cf. California shift, Canada shift

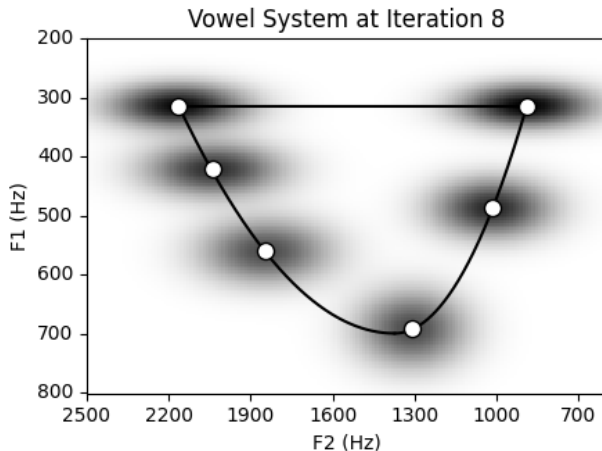


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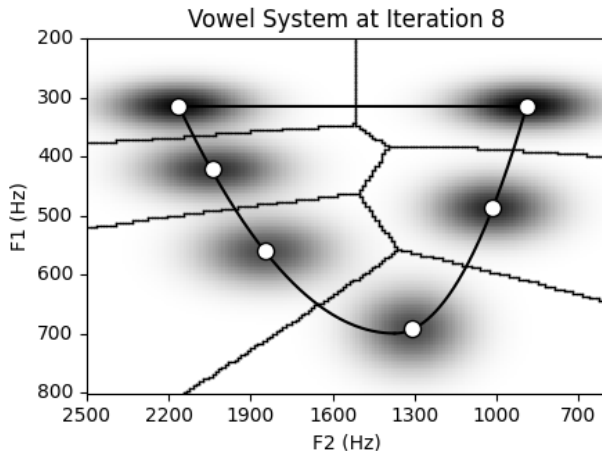


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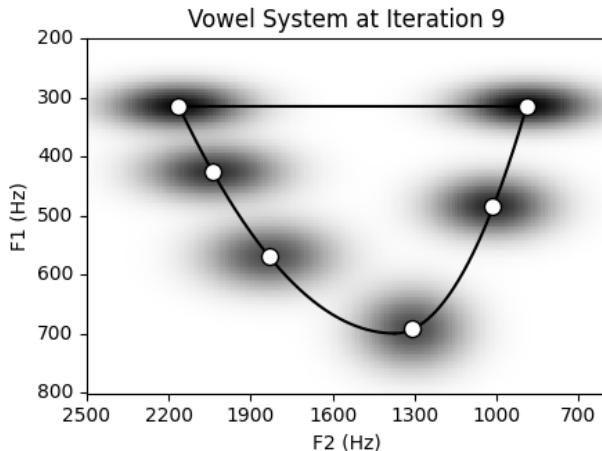


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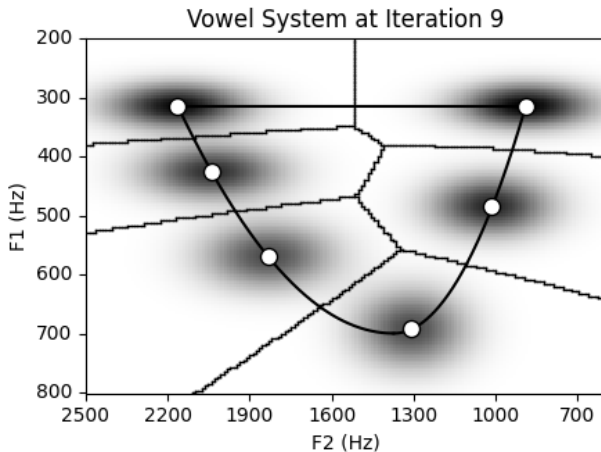


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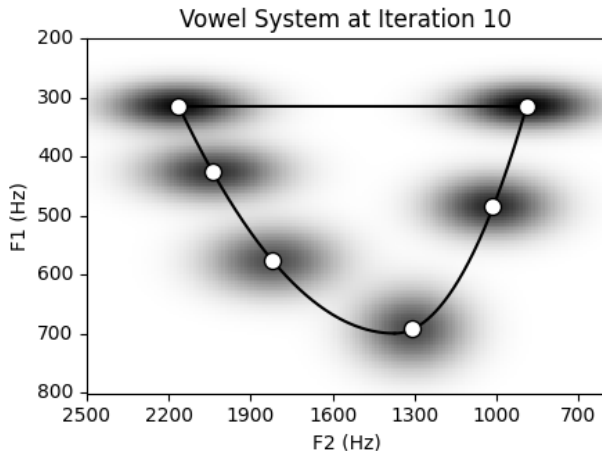


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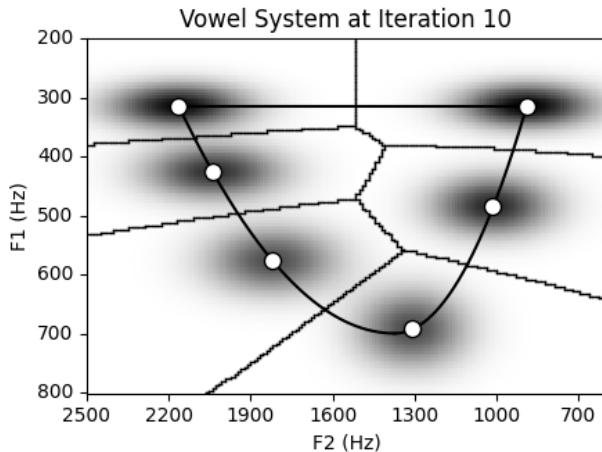


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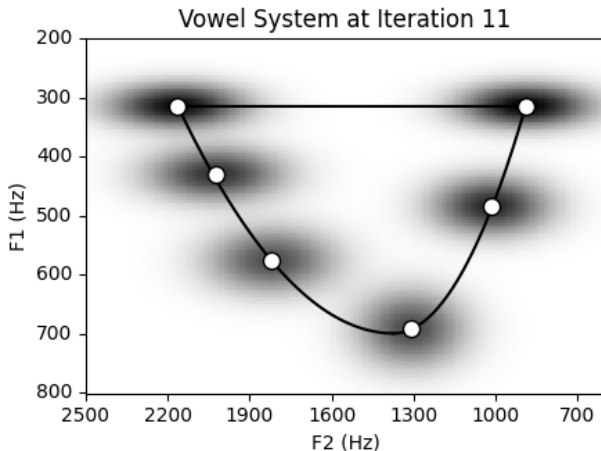


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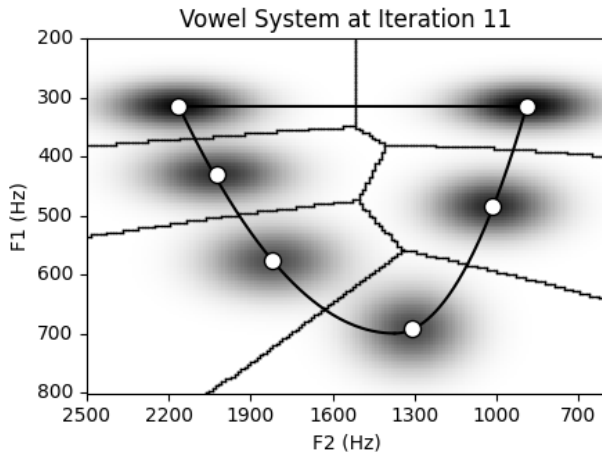


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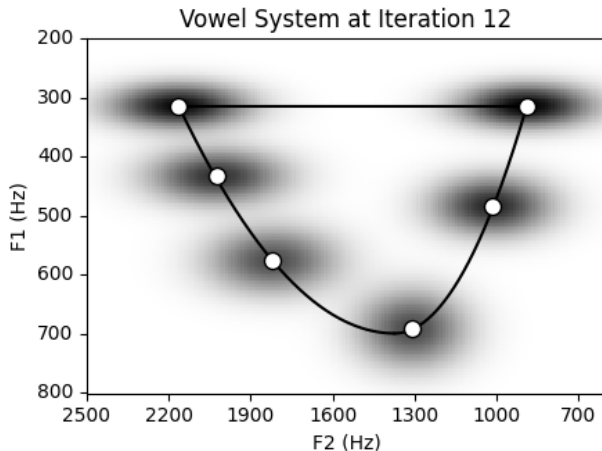


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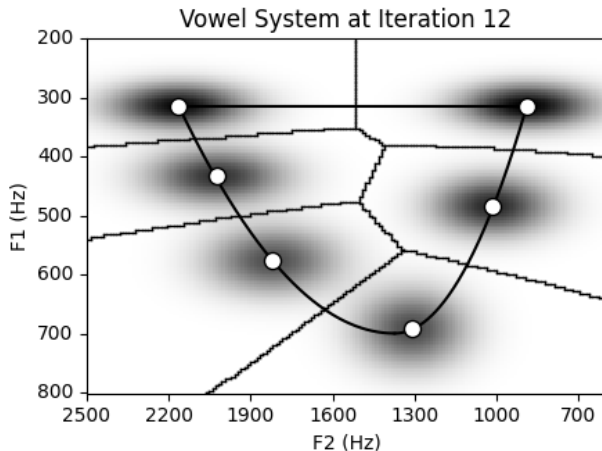
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ICM reaches equilibrium with naturalistic, dispersed vowel system



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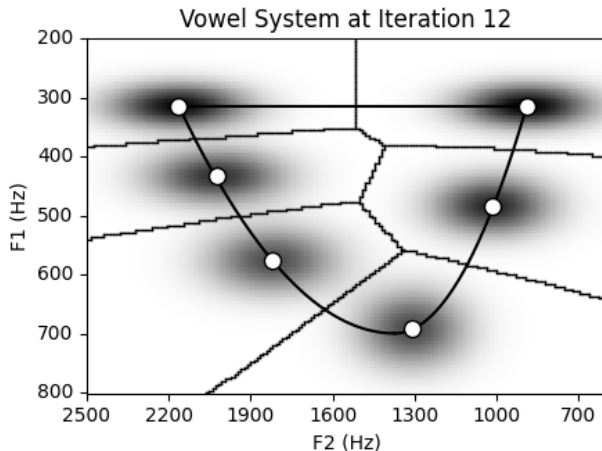
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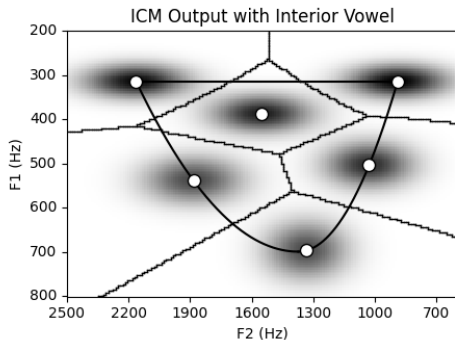
ICM reaches equilibrium with naturalistic, dispersed vowel system

- /ieɛaou/ 2nd most common 6V system (Schwartz et al., 1997a)



Example Result: Predicting Interior Vowels

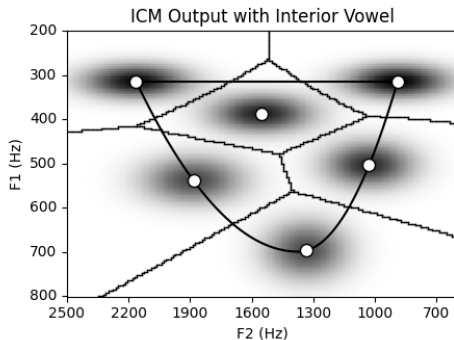
Most common 6V system /ieaouə/ (Schwartz et al., 1997a) also emerges



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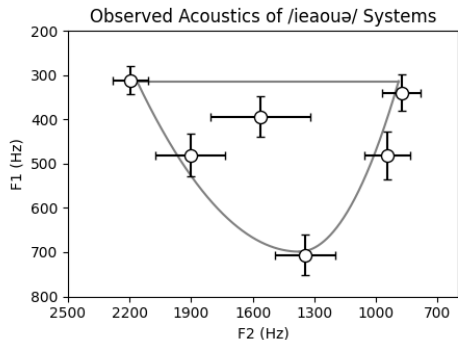
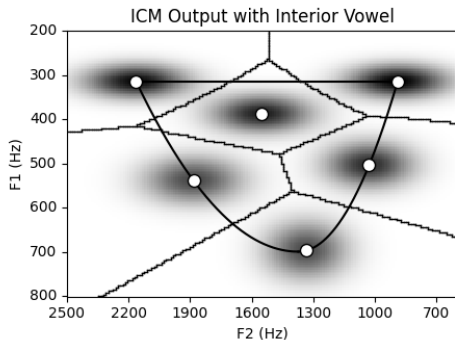
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Most common 6V system /ieaouə/ (Schwartz et al., 1997a) also emerges

- Improvement on dispersion models that fail to predict /ə/ (e.g. Schwartz et al, 1997b)
- Predicted acoustic value of /ə/ matches typology (Becker-Kristal, 2010)



Modeling Vowel System Typology

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Approach:

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- Run large number of ICM simulations over random initial systems

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Becker-Kristal (2010)'s Taxonomic Labels

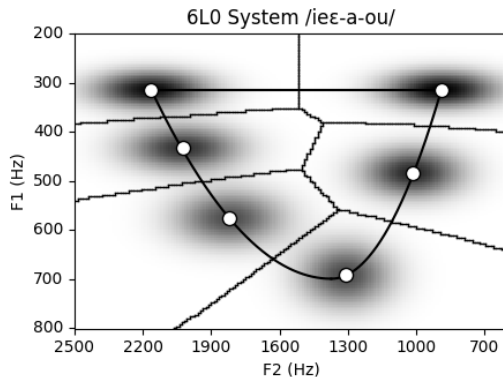
Systems take a label $\#_p X \#_{np}$, where:

- $\#_p$: number of peripheral vowels
- X indicates organization of periphery
 - S : symmetric
 - L : left-crowded (more front vowels)
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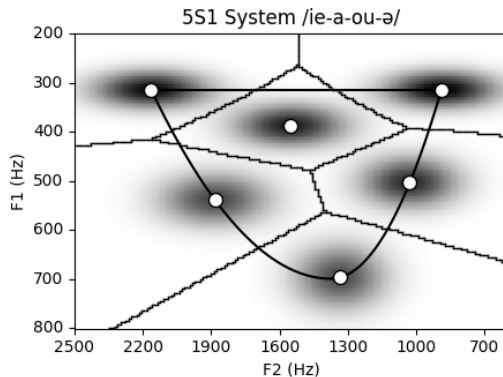
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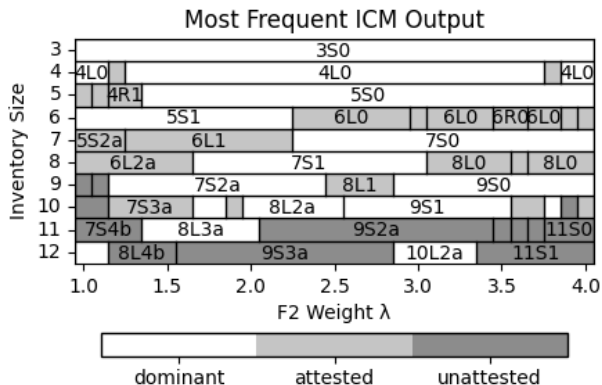


Typological Baseline

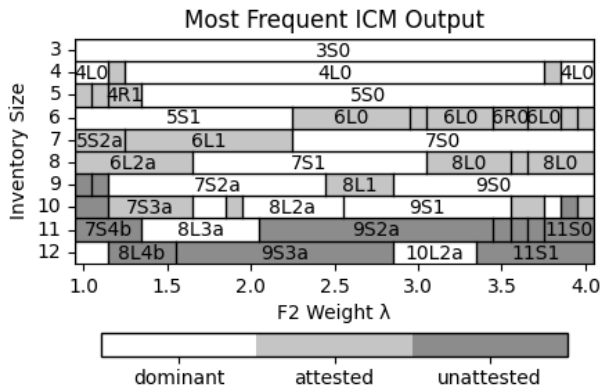
Most common system in UPSID by number of vowels:

Inventory Size	Dominant System	Example IPA	Frequency
3 (n=19)	3S0	/iaʊ/	1.000
4 (n=25)	4L0	/ieaʊ/	0.560
5 (n=109)	5S0	/ieaou/	0.890
6 (n=60)	5S1	/ieaouə/	0.683
7 (n=44)	7S0	/ieɛaou/	0.523
8 (n=19)	7S1	/ieɛaouə/	0.421
9 (n=24)	7S2a	/ieɛaouiə/	0.292
	9S0	/iieɛaouu/	0.292
10 (n=8)	8L2a	/iieɛaouiə/	0.250
	9S1	/iieɛaouuə/	0.250
11 (n=4)	8L3a	/ieɛæaouyøə/	0.667

Predicting Typologically Dominant Vowel Systems

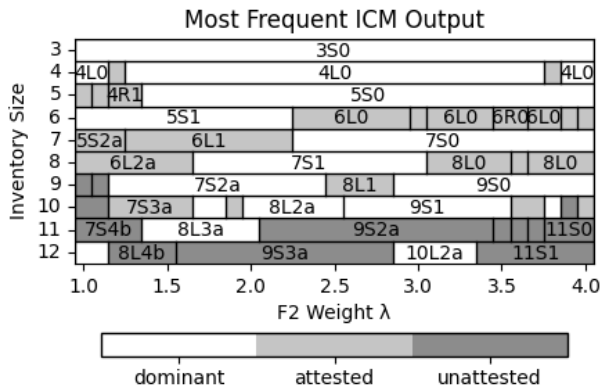


Predicting Typologically Dominant Vowel Systems



Every dominant system up to 12 vowels most frequent under some setting

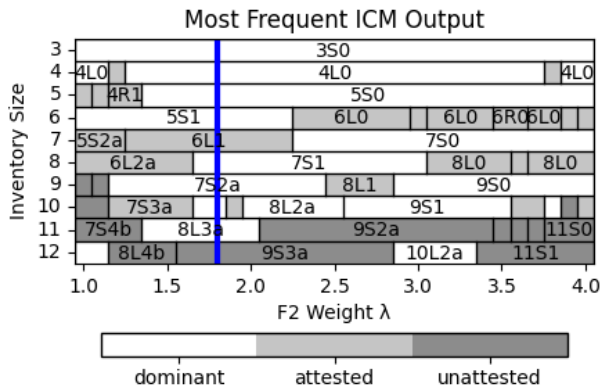
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inventory sizes > 7 not reported
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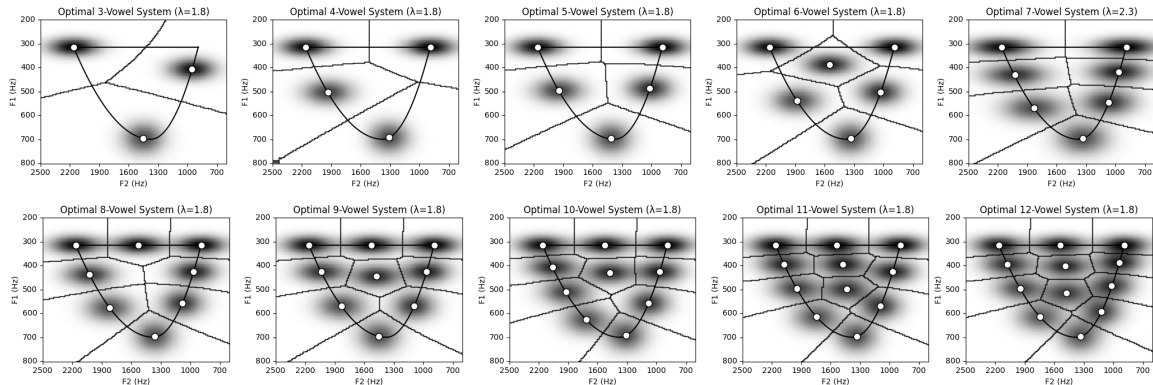


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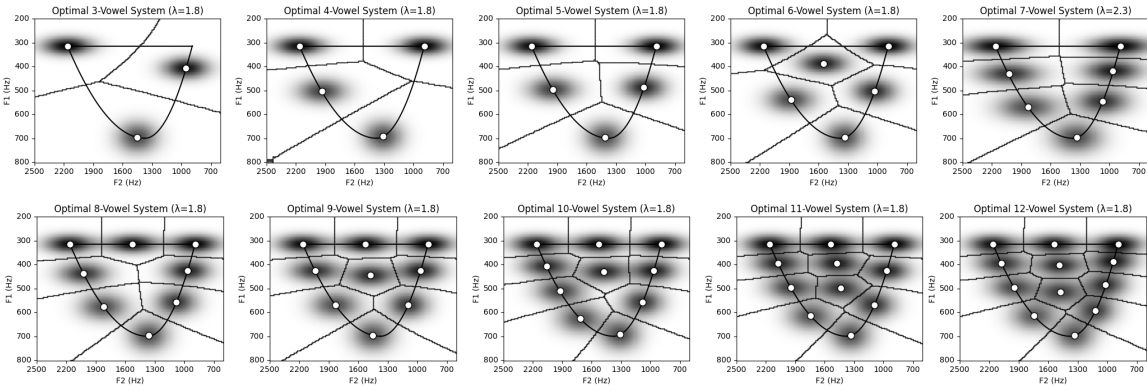
- Best model ($\lambda = 1.8$) predicts dominant system for 8/10 inventory sizes

Optimal Systems Visualized



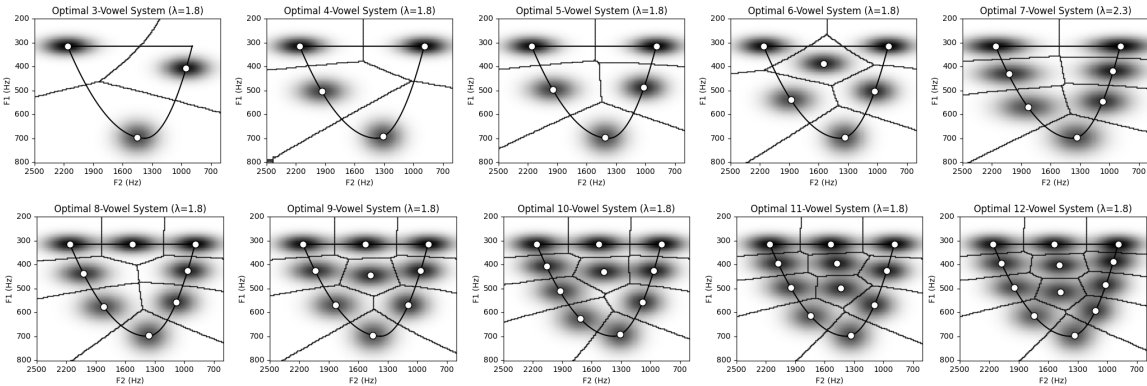
Optimal Systems Visualized

Optimal systems generally match acoustic typology (Becker-Kristal, 2010)



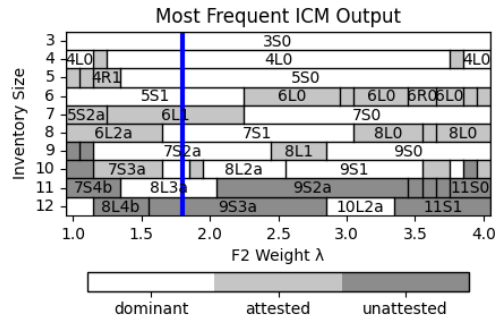
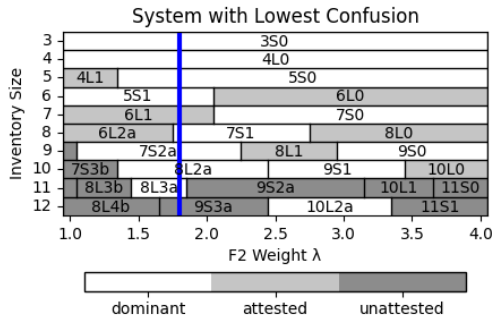
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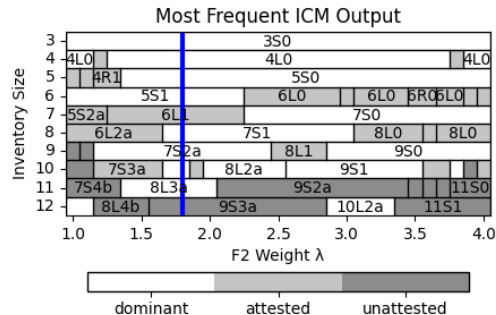
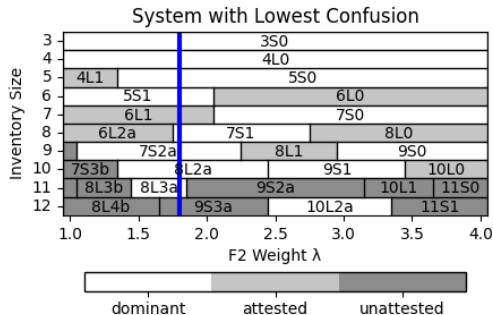
Improvement for larger inventories appears thanks to ability to predict interior vowels

Dominant Systems Optimize Confusion



Optimal system with lowest confusion tends to correspond to most frequent ICM output

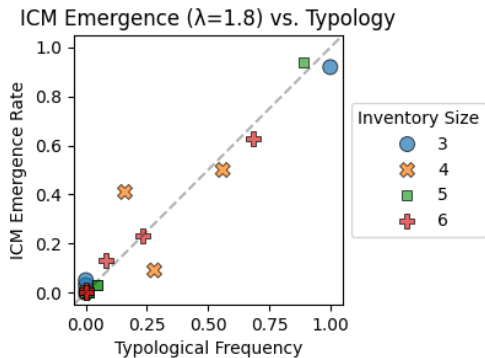
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 ⇒ Optimal systems common because realistic sound change mechanism produces them

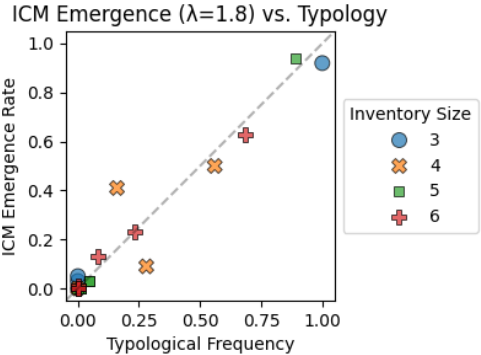
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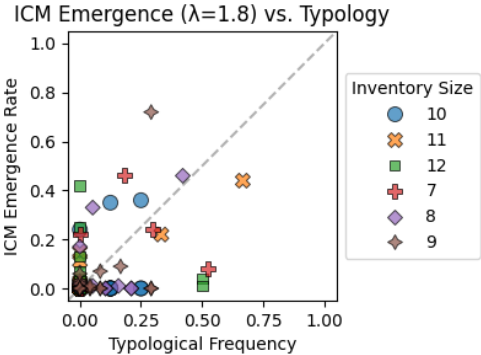


Tight correlation up to 6 vowels
(67% of languages)

Predicting Typological Frequencies



Tight correlation up to 6 vowels
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Less neat correlation from 7 vowels
(33% of languages)

Predicting Typological Frequencies

ICM matches or improves on de Boer (1999) up to 6 vowels

Inventory Size	Kullback-Leibler Divergence	
	de Boer (1999)	ICM ($\lambda = 1.8$)
3	0.248	0.084
4	∞	0.231
5	0.009	0.009
6	0.028	0.022
7	0.383	0.874

Takeaways

ICM improves on Dispersion Theory (/ə/, larger systems, variation, mechanism)

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Resolution to tension between frameworks emphasizing optimality of sound systems (Optimality Theory) vs. non-goal-oriented sound change (Evolutionary Phonology)

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- Short-sighted local optimizations can accumulate over time into globally optimal structure at level of sound systems

Future Directions

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- ICM over words predicts lack of dispersion between phonemes with few minimal pairs
 - Resolution to discrepancy between dispersion and ubiquity of phonemic merger?

- «[Back to top](#)»

Acknowledgements & Link to Paper

I would like to thank:

- Bruce Hayes, Megha Sundara, Meg Cychosz, Patricia Keating, Roy Becker-Kristal, the UCLA phonetics seminar, and three anonymous reviewers for their thoughtful comments
- Research assistants Valerie So, Shawn Grant, Remi Akopians, Bill Tran, Makenna Kumlue, and Christopher Yang for their help creating the vowel acoustics data set used to fit the formant standard deviation model
- UCLA Division of Graduate Education's Graduate Summer Research Mentorship program for partial funding



Link to Paper

Thank You!

Implementing ICM over the Vowel Space

boundaries on the vowel space

We will consider vowel distributions with F1 and F2 means (μ_1, μ_2) in Hz limited by the following equations, adapted from Liljencrants & Lindblom (1972) to match the space reported by Becker-Kristal (2010) for /ieaou/:

$$314 \leq \mu_1 \leq 699 \quad (1)$$

$$\mu_2 \leq 1373 + (2164 - 1373) \sqrt{\frac{699 - \mu_1}{699 - 314}} \quad (2)$$

$$\mu_2 \geq 1373 + (889 - 1373) \sqrt{\frac{699 - \mu_1}{699 - 314}} \quad (3)$$

N.B. – Simulations here use Hz (not psychoacoustic scales used in past models) since phonetic overlap does not depend on choice of scale.

Implementing ICM over the Vowel Space

set of possible vowel distributions

Vowels are modeled as bivariate normal distributions with 0 covariance:

$$(F1, F2) \sim \mathcal{N} \left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix} \right) \quad (4)$$

where:

- formant means (μ_1, μ_2) are limited by the previous equations, and
- standard deviations (σ_1, σ_2) are calculated as a function of formant means using a linear regression model fit to a crosslinguistic dataset.

Implementing ICM over the Vowel Space

formant standard deviation model

The standard deviation model was fit to z-scored data for 19 speakers of 21 languages from 10 studies^a with random effects for speaker and study:

$$z_1 = \frac{\mu_1 - 498.7212}{163.5568} \quad z_2 = \frac{\mu_2 - 1460.05}{543.4261} \quad (5)$$

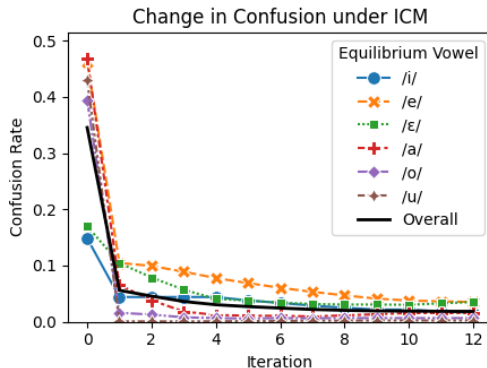
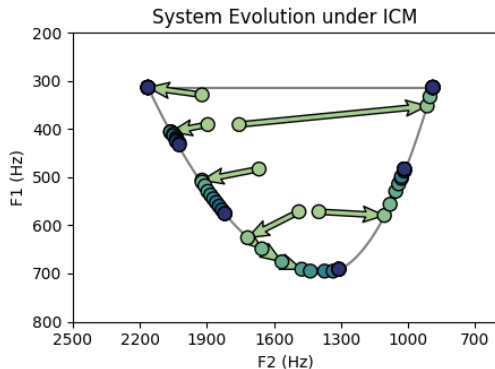
$$\sigma_1 = 34.9728 + 8.0738z_1 - 0.2633z_2 - 0.3906z_1z_2$$

$$\sigma_2 = \lambda(86.803 - 6.611z_1 + 6.741z_2 + 5.136z_1z_2)$$

^aHoltse, 1972; Thongkum, 1986; McCord, 1989; Svantesson, 1995; Olson, 2001; Calamai et al., 2003; Calamai, 2004; Thornell and Nagano-Madsen, 2004; Chiang and Chiang, 2005; Mayr and Davies, 2009

Following past dispersion models, this model also experiments with F2 scaling, implemented as a weight $1.0 \leq \lambda \leq 4.0$ on σ_2 .

Each ICM Iteration Reduces Overall Confusion



Kullback-Leibler Divergence

Kullback-Leibler (KL) divergence

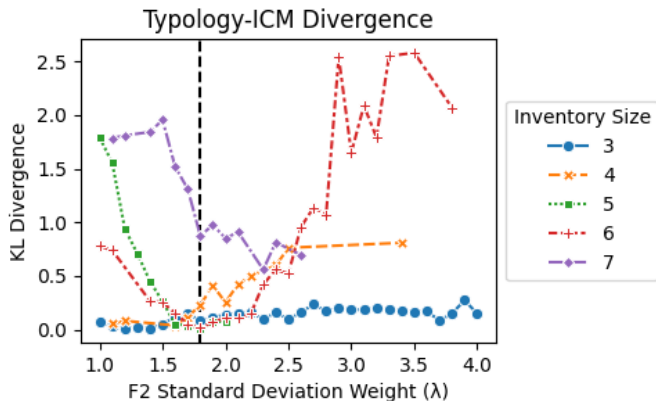
$$D_{KL}(P||Q) = \sum_S P(S) \log \left(\frac{P(S)}{Q(S)} \right)$$

where:

- P = typological frequencies
- Q = model emergence rates

Lower KL divergence = better model

KL Divergence across F2 Weights



Again, $\lambda = 1.8$ yields best overall model

Probabilistic Categorization

Actual human phonetic categorization is *not* perfectly categorical

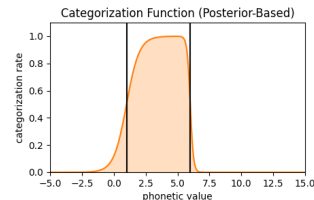
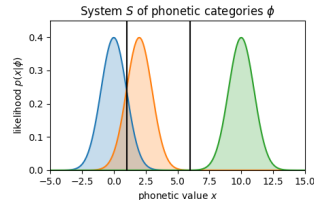
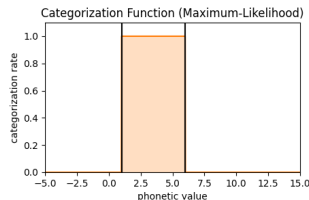
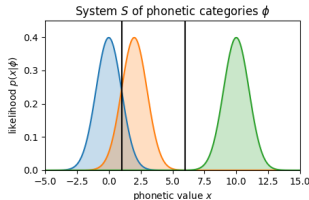
We can use posterior probabilities for probabilistic categorization

maximum-likelihood categorization

$$C(x) = \underset{\phi \in S}{\operatorname{argmax}} p(x|\phi)$$

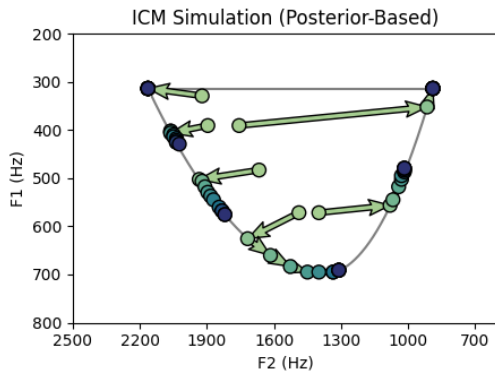
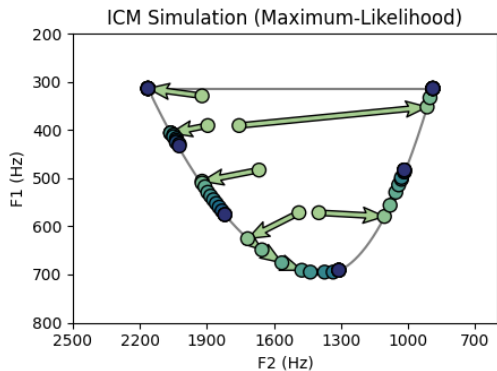
posterior-based categorization

$$P(C(x) = \phi) = \frac{p(x|\phi)P(\phi)}{\sum_{\phi' \in S} P(\phi')}$$



Categorical (Maximum-Likelihood) vs. Probabilistic (Posterior-Based) ICM

More realistic probabilistic listener behavior does not significantly alter behavior of ICM



ICM Over Biphone Sequences to Model Allophony

Sequences $\{/ki/, /ka/, /ku/, /qi/, /qa/, /qu/\}$ modeled as 3-dimensional normal distributions, combining vowel space model with idealized 1-dimensional $/k/-/q/$ continuum ($\mathcal{N}(-1, 1)$ and $\mathcal{N}(1, 1)$)

Result: allophonic split of $/iau/$ into $[iæu]$ after $/k/$ and $[eao]$ after $/q/$ (cf. Arabic, Greenlandic, Quechua)

